Humoral immunity to cow's milk proteins and gliadin within the etiology of recurrent aphthous ulcers?

OBJECTIVES: The goal of this study was to determine the incidence of serum antibodies to gliadin and to cow's milk proteins (CMP) using ELISA test, within patients who have recurrent aphthous ulcers (RAU). SUBJECTS AND METHODS: Fifty patients with recurrent aphthous ulcers and fifty healthy people were included in this research. Levels of serum IgA and IgG antibodies to gliadin and IgA, IgG and IgE to CMP were determined using ELISA. RESULTS: The levels of serum antigliadin IgA and IgG antibodies were not significantly higher in patients with RAU in comparison with the controls ($P = 0.937$ and $P = 0.1854$ respectively). The levels of serum anti-CMP IgA, IgG and IgE antibodies were significantly higher in patients with RAU in comparison with the controls ($P < 0.005$, $P < 0.002$ and $P < 0.001$ respectively). In general, the increased humoral (IgA or IgG) immunoreactivity to CMP was found in 32 of 50 patients, while 17 of them showed the increased levels of both IgA and IgG immunoreactivity to CMP. At the same time, 16 out of 50 patients had IgA, IgG and IgE immunoreactivity to CMP. CONCLUSION: These results indicate the strong association between high levels of serum anti-CMP IgA, IgG and IgE antibodies and clinical manifestations of recurrent aphthous ulcers.